

MA 621: Modeling Population Dynamics

Course Code	MA 621
Course Name	Modeling Population Dynamics
L-T-P-C	3-0-0-3
Prerequisites	MA 513 Ordinary Differential
Intended for	M.Sc./B.Tech/M.S./M.Tech/Ph.D.
Distribution	Elective
Approval	22nd Senate (Jun 2019)

Course Contents

- **Introduction** [2 Lectures]
Modeling Nature: History and General philosophy of the modeling approach. The demand for Reliable Prediction and Latest Development in Mathematical Modeling. Advantages and Demerits of Mathematical Modeling while Dealing with Real world's Problems
- **Unstructured Population Models** [8 lectures]
Population Dynamics from the first principle, Single-Species Models with Exponential and Logistic growth, Self Limitation, Consumer Resource Oscillation. Application of theory of Difference Equations to Population Growth models. Introduction to Discrete- time Models. Linear Models, Growth Models, Harvest Models: Bifurcation and Breakpoints. Delayed Differential models, Exogenous Drivers.
- **Introduction to Continuous Models** [10 lectures]
Models for Single Species Populations: Malthus Model, Logistic Growth, Allee Effect. Predator-Prey System and the Lotka-Volterra Equations, Populations in Competition, Multiple-Species Communities and the Routh-Hurwitz Criteria, Qualitative Stability Types of Models: Continuous-time, Discrete-time, Delayed Differential models, Exogenous Drivers.
- **Age and Stage-Structured Models In Ecology** [6 lectures]
Discrete Time Models with Age and Stage classes, Leslie Matrices, Estimating the Transition Matrix From Empirical Data, fisheries and Insect Populations as Prototype Example.
- **Population Interactions** [8 lectures]
Functional response, Aggregative response, Numerical Response, Competition Models, Mutualism Models, Lotka Volterra Model, Anatomy of Predator Prey Cycle, Grazing systems, pathogens and parasites, Tritrophic Models.
- **Empirical Approaches** [8 lectures]
Analysis of Population fluctuations, Time Series Analysis, Fitting Models to Data, Fitting Mechanistic Models

Text Books

1. Larry L. Rockwood, **Introduction to Population Ecology**, 2nd Edition, Blackwell Publishing Limited, 2015.

References

1. Richard Haberman, **Mathematical Models**, 5th edition, SIAM, 2013.
2. Leah Edelstein Keshet, **Mathematical Models in Biology**, SIAM, 2005
3. Peter Turchin, **Complex Population Dynamics: A Theoretical Empirical Synthesis**, Princeton University Press, 2003.
4. Jo Smith and Pete Smith, **Environmental Modelling: An Introduction**, Oxford University Press, 2007.
5. Mark Kot, **Elements Of Mathematical Ecology**, 1st Edition, Cambridge University Press, 2001.